

Mumbai Petrol Pump Sentiment Analysis

Methodology, Assumptions & Data Documentation

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1. Overview

This document records the methodology decisions, assumptions, and known data limitations applied during the expansion of the petrol pump sentiment dashboard to cover 163 stations across Greater Mumbai. Since the Mumbai dataset is structured differently from the Pune data — containing station-level aggregate ratings rather than individual review text — a number of deliberate methodology choices were made, all of which are documented here for transparency and traceability.

2. Data Sources

Dataset	Format	Description
Pune Petrol Pumps Pune_Retail_outlet.xlsx	Review-level XLSX	568 individual reviews with review text, star ratings, and timestamps. Primary dataset powering the BERT sentiment model.
Mumbai Petrol Pumps mu mbai_petrol_pumps.xlsx	Station-level XLSX	168 station records with aggregate Google ratings (totalScore), review counts, coordinates, and fuel category. No individual review text.
Mumbai Boundary mumbai_boundary.geojson	GeoJSON	A MultiPolygon (5 rings) defining the Greater Mumbai Municipal Corporation (BMC) boundary in WGS-84 coordinates. Used to draw the city outline on the geographic map and confirm all stations fall within the BMC jurisdiction.

3. Assumptions — Pune / Review-Level Analysis

A1 • BERT Sentiment Model

The `nlptown/bert-base-multilingual-uncased-sentiment` model from HuggingFace was used for sentiment classification. Trained on Amazon, Yelp, and TripAdvisor reviews across 6 languages, it is well-suited to the short Indian-English and code-switched reviews in the Pune dataset. The model outputs a 1–5 star prediction mapped as follows:

- Stars 1–2 → **Negative**
- Star 3 → **Neutral**
- Stars 4–5 → **Positive**

A2 • Negative Override Rules

The multilingual BERT model occasionally misclassifies complaint language in code-switched text. To correct for this, hard override rules were implemented: if a review contains phrases such as *fraud*, *scam*, *no cng*, *overcharg*, *ripoff*, or *theft*, the prediction is forced to Negative regardless of the model output.

A3 · Issue Classification

A keyword-based scoring system was built across 12 issue categories: Cleanliness, Staff Behaviour, Payment Issue, Waiting Time, Fuel Quality, Billing or Trust Issue, Safety Concern, Facility Maintenance, Customer Amenities, Traffic or Queue Management, Accessibility, and CNG Availability. Each review is assigned the category with the highest keyword match count. A secondary tag (top-2 categories) is also generated for reviews that touch multiple issues. Reviews with no keyword matches are labelled "Other".

A4 · Review Text Normalisation

Before passing text to the model, URLs are stripped, non-word characters are replaced with spaces, and text is truncated to 512 characters. Blank or null review texts are excluded from analysis entirely.

4. Assumptions — Mumbai / Station-Level Analysis

B1 · No Individual Review Text Available

The Mumbai dataset contains one row per petrol pump station with aggregate ratings — not individual reviews. Because the BERT model requires per-review text as input, it was not applied to this dataset. Sentiment is instead derived from the aggregate rating using threshold rules (see B2). As a result, Mumbai sentiment labels are statistical proxies, not the output of text analysis.

B2 · Sentiment Thresholds (Rating-Derived)

The following thresholds were defined, consistent with the BERT star-to-label mapping used for Pune:

Rating Band	Sentiment Label	Rationale
≤ 2.5 ★	Negative	Consistently poor customer experience
2.6 – 3.5 ★	Neutral	Mixed or average performance
> 3.5 ★	Positive	Generally satisfactory

B3 · Stations Excluded from Analysis

- 5 stations have a null *totalScore* and are labelled "Unknown". They are excluded from all sentiment charts and the scorecard but remain visible in the raw data view.
- 2 stations have null coordinates and are excluded from the geographic map only.

B4 · Geographic Zone Assignment

Each station was assigned to one of five geographic zones based on WGS-84 coordinate thresholds, approximating Mumbai's colloquial and administrative boundaries:

Zone	Latitude Range	Longitude	Areas Covered
South Mumbai	lat < 18.970	—	Colaba, Fort, Churchgate, Worli, Byculla
Central Mumbai	18.970 ≤ lat < 19.050	—	Dadar, Sion, Kurla, Chembur
Western Suburbs	19.050 ≤ lat < 19.150	lng < 72.880	Bandra, Juhu, Andheri, Versova
Eastern Suburbs	19.050 ≤ lat < 19.150	lng ≥ 72.880	Ghatkopar, Vikhroli, Mulund, Bhandup
North Suburbs	lat ≥ 19.150	—	Goregaon, Malad, Kandivali, Borivali

Note: These boundaries are coordinate-based approximations. For ward-level precision, a spatial join against Mumbai ward-level GeoJSON would be required.

B5 • Fuel Type Classification

Stations are classified as **CNG** if their `categoryName` value contains "compressed natural gas". All other stations — including those with a null `categoryName` — are classified as **Petrol / Diesel**, since the majority of Mumbai petrol pumps offer petrol and diesel as primary fuels.

B6 • CNG Availability Inference

CNG-categorised stations are treated as CNG-only facilities. Stations categorised as "Petrol Pump" may or may not offer CNG as an additional service — the dataset does not capture this distinction. Determining CNG availability at all stations would require cross-referencing OMC portals or a live fuel station API.

B7 • Low-Evidence Stations

Stations with fewer than 50 reviews are flagged as low-evidence in the Insights tab. Their sentiment labels carry less statistical weight; monitoring over time is recommended rather than immediate operational action based on limited data.

5. Validation Against Pune Baseline

Metric	Pune	Mumbai	Comparable?
Sentiment model	BERT on review text	Rating threshold	No — different methods
Issue categories	Keyword-NLP (12 categories)	Not available (no review text)	No
Time-series analysis	Available (review timestamps)	Not available (no timestamps)	No
Geographic map	Not exposed in UI	Full interactive map	—
Station comparison	Multi-outlet scorecard	Station scorecard	Yes — comparable

Key limitation: Pune and Mumbai sentiment results are not directly comparable because they use fundamentally different methods. A fair comparison would require either scraping individual reviews for all 168 Mumbai stations to run BERT, or deriving sentiment from `totalScore` for Pune as well.

6. Known Data Gaps

- 1. No per-review text for Mumbai stations** — restricts analysis to aggregate-level insights only and prevents NLP-based issue classification.
- 2. No timestamps** — the Mumbai file contains no review date or collection date per station, making time-series trend analysis impossible.
- 3. CNG availability at multi-fuel stations is unknown** — the dataset only identifies dedicated CNG stations, not petrol pumps that also offer CNG as an additional service.
- 4. Brand / OMC affiliation** (HPCL, BPCL, IOC, Shell, etc.) is not captured as a structured field — it could be extracted from station names via pattern matching but was not implemented to avoid over-engineering the current version.

5. Opening hours are present in the raw data but not parsed — a 24-hour vs limited-hours filter could be a useful addition in a future iteration.